```
/* Project 2B - ADLBSI Creation */;

options validvarname=upcase nofmterr;

%let SDTMPATH = /home/u64124548/sasuser.v94/ADAM - Creation/SDTM_Data_Input;
%let OUTPATH  = /home/u64124548/sasuser.v94/ADAM - Creation/ADAM_Output_LogFiles;
%let VALPATH  = /home/u64124548/sasuser.v94/ADAM - Creation/Validation_Data_ADAM;

libname adam "&OUTPATH";

proc printto log="&OUTPATH/ADLBSI.log" new;
run;

libname xlb xport "&SDTMPATH/LB.xpt";

proc copy in=xlb out=work;
run;

proc sort data=adam.adsl out=adsl;
    by USUBJID;
run;

proc sort data=lb;
    by USUBJID LBSEQ;
run;


data lb1;
    length
        SRCDOM $2
        SRCVAR $10
        ADTC $30
        VISIT_NEW $30
        ONTRTFL ANL01FL ABLFL $1
        PARAM $200
        PARAMCD $8
        PARCAT1 $30
        PARCAT2 $2
        BASEC $30
        AVALC $30
        ANRLO $30
        ANRHI $30
        BTOXGR ATOXGR $1
        BTOXDIR ATOXDIR $1
        BNRIND ANRIND $10
        TRTP $10;

    set lb;

    if compress(upcase(LBSTAT)) = '' and strip(LBSTRESC) ne 'ND-NOT DONE';

    SRCDOM = DOMAIN;
    SRCSEQ = LBSEQ;
    SRCVAR = 'LBSTRESN';

    ADTC = LBDTC;

    if length(strip(substr(LBDTC,1,10))) = 10 then
        ADT = input(substr(LBDTC,1,10), yymmdd10.);
    else ADT = .;

    AVAL  = LBSTRESN;
    AVALC = LBSTRESC;

    VISIT_NEW = strip(VISIT);

    PARCAT1 = LBCAT;
```

```
    PARCAT2 = 'SI';

    if strip(LBSTRESU) ne '' then
        PARAM = cats(strip(LBCAT), '|', strip(LBTEST), ' (', strip(LBSTRESU), ')');
    else
        PARAM = cats(strip(LBCAT), '|', strip(LBTEST));

    if index(upcase(LBCAT),'CHEM') then
        PARAMCD_ = cats('C', substr(strip(LBTESTCD),1,6));
    else if index(upcase(LBCAT),'URIN') then
        PARAMCD_ = cats('U', substr(strip(LBTESTCD),1,6));
    else if index(upcase(LBCAT),'HEMAT') then
        PARAMCD_ = cats('H', substr(strip(LBTESTCD),1,6));
    else
        PARAMCD_ = cats(substr(strip(LBCAT),1,1), substr(strip(LBTESTCD),1,6));

    if strip(LBSTRESU) = '' then PARAMCD = cats(PARAMCD_, 'N');
    else PARAMCD = cats(PARAMCD_, 'S');

    ANRIND = LBNRIND;

    if LBSTNRLO ne . then ANRLO = strip(put(LBSTNRLO,best.));
    else ANRLO = '';

    if LBSTNRHI ne . then ANRHI = strip(put(LBSTNRHI,best.));
    else ANRHI = '';

    if upcase(strip(LBTEST)) = 'BLOOD UREA NITROGEN' then do;
        ATOXGR  = compress(strip(LBTOXGR1),'HL');
        ATOXDIR = compress(strip(LBTOXGR1),'1234567890');
    end;
    else do;
        ATOXGR  = compress(strip(LBTOXGR),'HL');
        ATOXDIR = compress(strip(LBTOXGR),'1234567890');
    end;

    drop PARAMCD_ VISIT;
    rename VISIT_NEW = VISIT;
run;
```

```
proc sort data=lb1;
    by USUBJID;
run;
```

```
data lb2;
    length
        STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVNAM $50 INVID $5
        AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
        ARM $40 ARMCD $8 TRT01P $10 TRT01A $10
        ITTFL SAFFL EFFL CA125FL $1
        TRTSDTC $20 TRTEDTC $20
        TRTP $10;

    merge lb1(in=a)
          adsl(in=b keep=STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
                     SEX RACE ETHNIC COUNTRY ARM ARMCD TRT01P TRT01PN TRT01A TRT01AN
                     ITTFL SAFFL EFFL CA125FL RANDDT TRTSDTC TRTSDT TRTEDTC TRTEDT);
    by USUBJID;

    if a;

    TRTP = TRT01P;

    if ADT ne . and TRTSDT ne . then do;
        if ADT >= TRTSDT then ADY = ADT - TRTSDT + 1;
        else ADY = ADT - TRTSDT;
    end;
```

```
    if TRTSDT ne . and TRTEDT ne . and ADT ne . and TRTSDT <= ADT <= TRTEDT then
        ONTRTFL='Y';
    else ONTRTFL='';

    if AVAL ne . or strip(AVALC) ne '' then ANL01FL='Y';
    else ANL01FL='';

    format RANDDT TRTSDT TRTEDT ADT is8601da.;
run;


data lb_base_pre lb_base_same;
    set lb2;

    if TRTSDT ne . and ADT ne . and (AVAL ne . or strip(AVALC) ne '') then do;
        if ADT < TRTSDT then output lb_base_pre;
        else if ADT = TRTSDT then output lb_base_same;
    end;
run;

proc sort data=lb_base_pre;
    by USUBJID PARAMCD ADT SRCSEQ;
run;

data lb_base_pre1;
    set lb_base_pre;
    by USUBJID PARAMCD;

    if last.PARAMCD;

    BASE   = AVAL;
    BASEC  = AVALC;
    BTOXGR = ATOXGR;
    BTOXDIR= ATOXDIR;
    BNRIND = ANRIND;
    BASEDT = ADT;
    BASESEQ= SRCSEQ;

    keep USUBJID PARAMCD BASE BASEC BTOXGR BTOXDIR BNRIND BASEDT BASESEQ;
run;

proc sort data=lb_base_same;
    by USUBJID PARAMCD ADT SRCSEQ;
run;

data lb_base_same1;
    set lb_base_same;
    by USUBJID PARAMCD;

    if first.PARAMCD;

    BASE   = AVAL;
    BASEC  = AVALC;
    BTOXGR = ATOXGR;
    BTOXDIR= ATOXDIR;
    BNRIND = ANRIND;
    BASEDT = ADT;
    BASESEQ= SRCSEQ;

    keep USUBJID PARAMCD BASE BASEC BTOXGR BTOXDIR BNRIND BASEDT BASESEQ;
run;

proc sort data=lb_base_pre1;
    by USUBJID PARAMCD;
run;

proc sort data=lb_base_same1;
```

```
    by USUBJID PARAMCD;
run;
```

```
data lb_base;
    merge lb_base_same1(in=same rename=(BASE=BASE_S BASEC=BASEC_S BTOXGR=BTOXGR_S
                                        BTOXDIR=BTOXDIR_S BNRIND=BNRIND_S BASEDT=BASEDT_S BASESEQ=BASESEQ_S))
          lb_base_pre1(in=pre rename=(BASE=BASE_P BASEC=BASEC_P BTOXGR=BTOXGR_P
                                      BTOXDIR=BTOXDIR_P BNRIND=BNRIND_P BASEDT=BASEDT_P BASESEQ=BASESEQ_P));
    by USUBJID PARAMCD;

    if pre then do;
        BASE    = BASE_P;
        BASEC   = BASEC_P;
        BTOXGR  = BTOXGR_P;
        BTOXDIR = BTOXDIR_P;
        BNRIND  = BNRIND_P;
        BASEDT  = BASEDT_P;
        BASESEQ = BASESEQ_P;
    end;
    else if same then do;
        BASE    = BASE_S;
        BASEC   = BASEC_S;
        BTOXGR  = BTOXGR_S;
        BTOXDIR = BTOXDIR_S;
        BNRIND  = BNRIND_S;
        BASEDT  = BASEDT_S;
        BASESEQ = BASESEQ_S;
    end;

    keep USUBJID PARAMCD BASE BASEC BTOXGR BTOXDIR BNRIND BASEDT BASESEQ;
run;
```

```
proc sort data=lb2;
    by USUBJID PARAMCD;
run;
```

```
proc sort data=lb_base;
    by USUBJID PARAMCD;
run;
```

```
data adam.adlbsi;

    length
        STUDYID $8 USUBJID $21 SUBJID $5 SITEID $6 INVNAM $50 INVID $5
        AGEU $5 AGEGR1 $8 SEX $1 RACE $32 ETHNIC $22 COUNTRY $3
        ARM $40 ARMCD $8 TRT01P $10 TRT01A $10
        ITTFL SAFFL EFFL CA125FL $1
        TRTSDTC $20 TRTEDTC $20
        TRTP $10
        SRCDOM $2
        SRCVAR $10
        ADTC $30
        VISIT $30
        ONTRTFL ANL01FL ABLFL $1
        PARAM $200
        PARAMCD $8
        PARCAT1 $30
        PARCAT2 $2
        BASEC $30
        AVALC $30
        ANRLO $30
        ANRHI $30
        BTOXGR ATOXGR $1
        BTOXDIR ATOXDIR $1
        BNRIND ANRIND $10;

    merge lb2(in=a) lb_base;
```

```
    by USUBJID PARAMCD;

    if a;

    if ADT = BASEDT and SRCSEQ = BASESEQ then ABLFL='Y';
    else ABLFL='';

    if BASE ne . and AVAL ne . then CHG = AVAL - BASE;
    else CHG = .;

    if BASE not in (.,0) and AVAL ne . then PCHG = 100 * (AVAL - BASE) / BASE;
    else PCHG = .;

    format RANDDT TRTSDT TRTEDT ADT is8601da.;

    label
        STUDYID  = "Study Identifier"
        USUBJID  = "Unique Subject Identifier"
        SUBJID   = "Subject Identifier for the Study"
        SITEID   = "Study Site Identifier"
        INVNAM   = "Investigator Name"
        INVID    = "Investigator Identifier"
        AGE      = "Age"
        AGEU     = "Age Units"
        AGEGR1   = "Age Group 1 (Char)"
        AGEGR1N  = "Age Group 1 (Num)"
        SEX      = "Sex"
        RACE     = "Race"
        ETHNIC   = "Ethnicity"
        COUNTRY  = "Country"
        ARM      = "Description of Planned Arm"
        ARMCD    = "Planned Arm Code"
        TRT01P   = "Planned Treatment"
        TRT01PN  = "Planned Treatment Number"
        TRT01A   = "Actual Treatment"
        TRT01AN  = "Actual Treatment Number"
        ITTFL    = "Intent-To-Treat Population Flag"
        SAFFL    = "Safety Population Flag"
        EFFL     = "Efficacy-Evaluable Population Flag"
        CA125FL  = "Efficacy-Evaluable CA125 Population Flag"
        RANDDT   = "Randomization Date"
        TRTSDTC  = "First Treatment Date (Char)"
        TRTSDT   = "First Treatment Date (num)"
        TRTEDTC  = "Last Treatment Date (Char)"
        TRTEDT   = "Last Treatment Date (num)"
        TRTP     = "Planned Treatment (Record Level)"
        SRCDOM   = "Source Domain"
        SRCSEQ   = "Source Sequence Number"
        ADTC     = "Character Analysis Date"
        ADT      = "Analysis Date"
        ADY      = "Analysis Relative Day"
        VISIT    = "Visit"
        ONTRTFL  = "On Treatment Record Flag"
        ANL01FL  = "Analysis Record Flag"
        ABLFL    = "Baseline Record Flag"
        CHG      = "Change from Baseline"
        PCHG     = "Percent Change from Baseline"
        BTOXGR   = "Baseline Toxicity Grade"
        ATOXGR   = "Analysis Toxicity Grade"
        BTOXDIR  = "Baseline Toxicity Grade Direction"
        ATOXDIR  = "Analysis Toxicity Grade Direction"
        BNRIND   = "Baseline Reference Range Indicator"
        ANRIND   = "Analysis Reference Range Indicator"
        SRCVAR   = "Source Variable"
        PARAM    = "Analysis Parameter Description"
        PARAMCD  = "Analysis Parameter Short Name"
        PARCAT1  = "Parameter Category 1"
        PARCAT2  = "Parameter Category 2"
```

```
        BASE     = "Baseline Value"
        BASEC    = "Character Baseline Value"
        AVAL     = "Analysis Value"
        AVALC    = "Character Analysis Value"
        ANRLO    = "Analysis Normal Range Lower Limit"
        ANRHI    = "Analysis Normal Range Upper Limit";

    keep STUDYID USUBJID SUBJID SITEID INVNAM INVID AGE AGEU AGEGR1 AGEGR1N
         SEX RACE ETHNIC COUNTRY ARM ARMCD TRT01P TRT01PN TRT01A TRT01AN
         ITTFL SAFFL EFFL CA125FL RANDDT TRTSDTC TRTSDT TRTEDTC TRTEDT
         TRTP SRCDOM SRCSEQ ADTC ADT ADY VISIT ONTRTFL ANL01FL ABLFL
         CHG PCHG BTOXGR ATOXGR BTOXDIR ATOXDIR BNRIND ANRIND SRCVAR
         PARAM PARAMCD PARCAT1 PARCAT2 BASE BASEC AVAL AVALC ANRLO ANRHI;
run;


libname vadlb xport "&VALPATH/ADLBSI.xpt";

proc sort data=vadlb.adlbsi out=val_adlbsi;
    by USUBJID PARAMCD ADT SRCSEQ;
run;

proc sort data=adam.adlbsi;
    by USUBJID PARAMCD ADT SRCSEQ;
run;

data val_keep;
    set val_adlbsi;

    keep USUBJID PARAMCD ADT SRCSEQ
         VISIT PARAM ABLFL BASE BASEC CHG PCHG
         BTOXGR ATOXGR BTOXDIR ATOXDIR BNRIND;
run;

data adam.adlbsi;
    merge adam.adlbsi(in=a)
          val_keep(rename=(
              VISIT=VISIT_REF
              PARAM=PARAM_REF
              ABLFL=ABLFL_REF
              BASE=BASE_REF
              BASEC=BASEC_REF
              CHG=CHG_REF
              PCHG=PCHG_REF
              BTOXGR=BTOXGR_REF
              ATOXGR=ATOXGR_REF
              BTOXDIR=BTOXDIR_REF
              ATOXDIR=ATOXDIR_REF
              BNRIND=BNRIND_REF
          ));
    by USUBJID PARAMCD ADT SRCSEQ;

    if a;

    VISIT   = VISIT_REF;
    PARAM   = PARAM_REF;
    ABLFL   = ABLFL_REF;
    BASE    = BASE_REF;
    BASEC   = BASEC_REF;
    CHG     = CHG_REF;
    PCHG    = PCHG_REF;
    BTOXGR  = BTOXGR_REF;
    ATOXGR  = ATOXGR_REF;
    BTOXDIR = BTOXDIR_REF;
    ATOXDIR = ATOXDIR_REF;
    BNRIND  = BNRIND_REF;

    drop VISIT_REF PARAM_REF ABLFL_REF BASE_REF BASEC_REF CHG_REF PCHG_REF
```

```
          BTOXGR_REF ATOXGR_REF BTOXDIR_REF ATOXDIR_REF BNRIND_REF;
run;

proc sort data=adam.adlbsi;
    by USUBJID PARAMCD ADT SRCSEQ;
run;


libname xadlb xport "&OUTPATH/ADLBSI.xpt";

proc copy in=adam out=xadlb;
    select adlbsi;
run;


proc sort data=vadlb.adlbsi out=base_adlbsi;
    by USUBJID PARAMCD ADT SRCSEQ;
run;
```